DEPARTMENT OF COMPUTER SCIENCE & ENGINEERING
UNIVERSITY COLLEGE OF ENGINEERING (A)
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CLASS TEST-II

Subject: COMPUTER ORGANIZATION AND MICROPROCESSORS (AI402PC)

Class: B.E (AIML)
Semester: IV
Academic Year: 2024-25

Max Marks: 30
Duration: 1 hour
Date: 28-06-2025

	PART A Answer All Questions (3 x 2 = 6 M)	Marks	BT	CO
1. a)	Explain the importance of PUSH & POP instruction.	2	2	3
b)	Define Flag register in 8085 Microprocessor.	2	1	4
c)	Examine the importance of 8259 programmable interrupt controller.	2	4	4
d)	Analyze the function and arrangement of individual cells in the RAM structure of the 8051 microcontroller.	2	4	5
e)	List the difference between microprocessor and microcontrollers.	2	1	5
	PART B Answer Any Two Questions (2 x 7 = 14 M)			
2.	Explain the structure and functioning of cache memory with relevant terms like hit ratio and miss.	5	2	3
	Discuss the stages of instruction pipeline in RISC architecture with example.	5	4	
3.	Explain the architecture and function of the Intel 8257 DMA controller.	5	2	4
	Explain how the Programmable Interrupt Controller (8259A) works and its significance.	5		
4.	Explain the process of interfacing an ADC with a microcontroller, highlighting the necessary circuit connections and programming steps.	5	4	5
5	Describe the procedure for interfacing an LCD with an 8051 microcontroller, explaining the control signals and data transfer mechanism.	5	3	